PROJECT 30 | REVIEW 2 (WEEK 11)

Sports Club Membership & Tournament Management System

Normalized Database Design up to 3NF, DDL Implementation & Query Architecture

Review 2 Objectives & Scope

Key deliverables and implementation progress achieved for Week 11 milestone.

Schema Normalization

Refined relational schema up to Third Normal Form (3NF) eliminating update and deletion anomalies.

Constraint Enforcement

Implemented strict DDL constraints for scheduling overlaps, active memberships, and capacity limits.

Advanced SQL Suite

Developed complex JOINs, aggregate reports, nested subqueries, and operational database views.

Normalized Relational Schema (3NF)

Core relational tables structured to ensure atomicity and eliminate transitive functional dependencies.

Core Entities

Members: MemberID, Name, Contact, DOB

Plans: PlanID, Title, Duration, Fee

Sports: SportID, SportName, Category

Coaches: CoachID, Name, Specialization

Operations & Facilities

Facilities: FacilityID, Location, Capacity

Teams: TeamID, SportID, CoachID, Name

Payments: PaymentID, MemberID, PlanID, Date

Equipment: EquipmentID, ItemName, Status

Schedules & Results

Training: SessionID, TeamID, FacilityID, Time

Tournaments: TournamentID, Title, Dates

Fixtures: FixtureID, Team1, Team2, Venue

Results: ResultID, FixtureID, Score1, Score2

Integrity Constraints & Business Rules

Enforcing club regulations and operational safety via advanced SQL constraints.

No Overlapping Facility / Team Schedules

Implemented unique compound keys and overlapping interval exclusion constraints on booking times.

Active Membership Validation

CHECK constraints and foreign key linkages ensuring participation is restricted to unexpired membership records.

Equipment Limits & Facility Capacity

CHECK constraints restricting equipment issue count per member and capping training headcounts against facility limits.

Advanced SQL Queries & Database Views

Key operational reports generated through multi-table joins and aggregation.

Fixtures & Standings

Dynamic tournament standings calculated using aggregate wins, losses, and point differentials across teams.

View: vw_tournament_standings

Fee Dues & Payments

Identifies members with pending payment dues by comparing active plan costs against recorded payment transactions.

View: vw_pending_membership_dues

Equipment Status

Tracks gear allocation history, overdue returns, and current inventory stock status across sports divisions.

View: vw_active_equipment_issues

REVIEW 2 SUMMARY

Database Prototype Ready for Application Integration

The normalized database backend, integrity constraints, and initial mock dataset are fully verified. Proceeding to Review 3 frontend connectivity and CRUD development.

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Normalized Tables

3NF
Schema Compliance

100%
Constraint Validation